Species-Aware Review: Prokaryotic Molybdenum Cofactor (Moco) Biosynthesis in *Pseudomonas putida* KT2440

Module: GTP → cyclic pyranopterin monophosphate (cPMP) → molybdopterin (MPT) → Mo-MPT, with optional MGD/MCD dinucleotide variants **Target taxon:** *Pseudomonas putida* KT2440 (PSEPK; NCBI 160488; proteome UP000000556) **Assigned local bucket:** kegg:ppu00790 ("Folate biosynthesis") — *see boundary caveat below* **Scope of this review:** module satisfiability + gene-annotation curation, not a generic pathway essay.

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1. Executive summary

P. putida KT2440 **encodes a complete, satisfiable molybdenum cofactor biosynthesis module** from GTP to Mo-MPT, and it supports **both** optional dinucleotide branches (Mo-MGD and Mo-MCD). Every mechanistic step of the generic module maps to at least one credible KT2440 locus with UniProt/InterPro support, and the MCD branch is backed by **direct functional evidence** in the target strain (heterologous Mo-MCD molybdoenzymes mature to catalytically competent form in KT2440; P 12730200).

The main issues are **curation/metadata artifacts, not biological gaps**:

1. **Two essential enzymes are missing from the candidate list** — `moaD` (PP_1293) and `moeB` (PP_0735) — because they partition to the KEGG sulfur-relay map (ppu04122) instead of the folate bucket used to build the list. They are present in the genome and must be added.
2. **No dedicated `mogA`.** MPT adenylylation is supplied by the **MoaB paralog lineage** (`moaB-I` PP_2122 / `moaB-II` PP_4600), the lineage-specific alternative that the generic module explicitly permits.

3. **Paralog ambiguity / likely over-propagation.** Three "MoaA" and three "MobA-like/MoaB" paralogs exist; InterPro signatures cleanly separate the canonical enzyme from accessory paralogs.
4. **The assigned bucket ppu00790 is the wrong boundary** for this module; the true Moco genes live in ppu04122 + `moeA` / `mobA`.

Bottom line for curation: mark the core module **covered**; add MoaD/MoeB; mark MPT-adenylylation **covered-by-alternative (MoaB)**; mark MCD **candidate_uncertain→covered** pending a `mocA` naming decision; exclude the folate/riboflavin/queuosine genes from this module.

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2. Target-organism pathway definition

Included process (this module): construction of the pyranopterin dithiolene ligand from GTP, its dithiolene sulfuration to molybdopterin, molybdate insertion to Mo-MPT, and optional appending of GMP (→ MGD) or CMP (→ MCD).

Explicitly excluded (kept as neighboring processes): - **Molybdate uptake** — `modABC` (PP_2489-type; UniProt `modA` Q88G97, `modB` Q88G96, `modC` Q88G95) and regulator `modR` (Q88QX4). Transport, not synthesis. - **Terminal cofactor sulfuration / apoenzyme insertion chaperones** — e.g. XdhC-family accessory factors (PP_2480 Q88K13, PP_4231 Q88F67, `xdhC` PP_4280 Q88F19), `fdhD` sulfur carrier (PP_0257 Q88R77). These load/insert the finished cofactor into client enzymes. - **Mature molybdoenzyme reactions** — xanthine dehydrogenase (`xdhAB` PP_4279/…), `paoC` (Q88HP3), nicotinate dehydrogenase (`nicB` Q88FX8), assimilatory nitrate/sulfite reductase (Q88M71), formate dehydrogenases, MsrP (Q88DZ2), isoquinoline oxidoreductase β-subunits.

Database naming / boundary caveats: - The genuine Moco genes carry **KEGG map ppu04122 ("Sulfur relay system")** as primary bucket (`moaA, moaB-I/II, moaC, moaE, moaD, moeB` + `moaA` paralogs), *not* the assigned **ppu00790 ("Folate biosynthesis")**. Only the maturation enzymes `moeA` and `mobA` and the MobA-like paralogs sit in ppu00790. - KEGG also groups Moco under **map00790 (Folate)** and **module M00880 (Molybdenum cofactor biosynthesis)**; MetaCyc PWY-5963/PWY-6823 are the corresponding reference pathways. The four GTP-consuming pterin/deazaguanine pathways (folate, riboflavin, queuosine, Moco) co-occur in the candidate list **only because they share GTP as precursor** — they are separate modules and should stay separate.

3. Expected step model (generic module → KT2440 realization)

Step	Generic player	KT2440 gene(s)	Status
1a. Radical-SAM GTP 3′,8-cyclization	MoaA	`moaA` **PP_4597** (Q88E69, EC 4.1.99.22)	covered
1b. cPMP synthesis	MoaC	`moaC` **PP_1292** (Q88NC0, EC 4.6.1.17)	covered
2a. MoaD adenylylation (activation)	MoeB	`moeB` **PP_0735** (Q88PW3, EC 2.7.7.80)	covered *(missing from candidate list)*
2b. Sulfur carrier	MoaD	`moaD` **PP_1293** (Q88NB9)	covered *(missing from candidate list)*
2c. MPT synthesis	$MoaD_2$-$MoaE_2$ synthase	`moaE` **PP_1294** (Q88NB8, EC 2.8.1.12) + MoaD	covered
3a. MPT adenylylation	MogA **or** MoaB	`moaB-I` **PP_2122** / `moaB-II` **PP_4600** (no MogA present)	covered-by-alternative (MoaB)
3b. Molybdate insertion → Mo-MPT	MoeA	`moeA` **PP_2123** (Q88L14, EC 2.10.1.1)	covered
4a. Mo-MGD (optional)	MobA	`mobA` **PP_3457** (Q88HA3, EC 2.7.7.77)	covered
4b. Mo-MCD (optional)	MocA	**PP_2483** (Q88K10) / **PP_4230** (Q88F68), MobA-like	candidate_uncertain (functional evidence supports presence)

Gene order confirms functional clustering: `moaC-moaD-moaE` form a mini-operon (PP_1292-1294); `moaB-I`–`moeA` are adjacent (PP_2122-2123); `moaA`–`moaB-II` are near-adjacent (PP_4597/PP_4600). `moeB` is standalone (PP_0735).

4. Candidate genes and evidence

High-confidence core enzymes (EC-level UniProt annotation; treat as covered): - **`moaA` PP_4597 (Q88E69)** — GTP 3′,8-cyclase, EC 4.1.99.22. Radical-SAM (PF04055) + MoaA C-term (PF06463) + **MoaA-specific IPR013483**. The canonical, housekeeping MoaA. - **`moaC` PP_1292 (Q88NC0)** — cPMP synthase, EC 4.6.1.17. Unambiguous. - **`moaD` PP_1293 (Q88NB9)** — MPT-synthase sulfur carrier; ThiS/MoaD fold (PF02597, MoaD-specific IPR010038). *Absent from candidate list.* - **`moaE` PP_1294 (Q88NB8)** — MPT synthase catalytic subunit, EC 2.8.1.12. - **`moeB` PP_0735 (Q88PW3)** — MPT-synthase adenylyltransferase, EC 2.7.7.80; UniProt function: "adenylation by ATP of the C-terminal glycine of MoaD." ThiF/MoeB (PF00899). *Absent from candidate list.* - **`moeA` PP_2123 (Q88L14)** — MPT molybdotransferase, EC 2.10.1.1. - **`mobA` PP_3457 (Q88HA3)** — Mo cofactor guanylyltransferase, EC 2.7.7.77; **MGD-specific IPR013482**. Canonical MGD synthase.

MoaB paralogs (MPT-adenylylation / MogA surrogate): - **`moaB-I` PP_2122 (Q88L15)** and **`moaB-II` PP_4600 (Q88E67)** — MogA/MoaB MoCF_biosynth domain (PF00994, IPR001453). In many bacteria and archaea a catalytically competent MoaB performs the MogA (MPT-adenylyltransferase) reaction; genomic adjacency to `moeA` and `moaA` respectively supports functional partnership. Evidence type: **homology** + **operon context** (no direct KT2440 assay).

MobA-like paralogs (MCD/MocA candidates): - **PP_2483 (Q88K10)** and **PP_4230 (Q88F68)** — generic MobA-like NTP-transferase (PF12804/IPR025877) **lacking** the MobA/MGD signature IPR013482, the architecture expected for **MocA** cytidylyltransferases. PP_2483 sits in a Mo-hydroxylase maturation island (Ior β PP_2478, XdhC PP_2480, MoaA-paralog PP_2482); PP_4230 is near an XdhC/`xdhBC` cluster. Evidence type: **domain architecture** + **genomic context**, plus strain-level functional support (§ below). - **Quantitative caveat (this study):** global % identity to *E. coli* references cannot discriminate MGD from MCD here. `mobA` PP_3457 is clearly MobA/MGD (39.8% to *E. coli* MobA P32173 vs 32.5% to MocA Q46810), but PP_2483 (35.6% MobA / 33.9% MocA) and PP_4230 (34.0% MobA / 35.6% MocA) are essentially equidistant — consistent with the fact that *E. coli* MobA and MocA share only **36.5%** identity with each other. **Conclusion: sequence identity alone does not assign MGD vs MCD** for the two paralogs; genomic context is the better discriminator and a CTP-vs-GTP substrate assay is the decisive experiment. Both remain MocA candidates (PP_2483 favored on context, but not by sequence).

MoaA paralogs (likely over-propagation): - **PP_1969 (Q88LG4)** and **PP_2482 (Q88K11)** — radical-SAM (PF04055/PF06463) but **lacking MoaA-specific IPR013483** and any EC number. Likely non-canonical radical-SAM enzymes over-labeled "MoaA."

Direct target-strain functional evidence (MCD branch): - Quinaldine 4-oxidase Qox (Mo-MCD enzyme) was *"synthesized in a catalytically fully competent form by a heterologous host, P. putida KT2440 pKP1"* (P 12730200) → KT2440 has a working Mo-MCD maturation apparatus. - Caveats/nuance: heterologous Qor (P 12654012) was Mo-deficient and Ior (P 12023088) inactive in KT2440, indicating **client-specific insertion/chaperone requirements** downstream of Moco synthesis — an *apoenzyme-loading* limitation, outside this module's scope. A 2026 study again uses KT2440 to produce a molybdenum-containing xanthine oxidase where the accessory **XodC** aids maturation (P 41933999).

Not part of this module but present (neighbors): `modABC`/`modR` (molybdate transport), XdhC-family and `fdhD` (cofactor insertion/sulfur handling), and numerous molybdoenzymes — all correctly excluded.

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5. Gaps, ambiguities, and likely over-annotations

- **Metadata gap (not biological):** `moaD` (PP_1293) and `moeB` (PP_0735) are genuine, essential module members omitted from the candidate list due to KEGG bucket partitioning (ppu04122 vs ppu00790). **Action: add both.**
- **No `mogA` gene** in UP000000556 (searched by gene name and protein name). The MPT-adenylylation step is **covered by the MoaB alternative**, not a gap. Do **not** mark this step a gap for lack of MogA.
- **`mocA` naming gap:** MCD is made by MobA-like paralogs (PP_2483 best candidate) that are not literally named `mocA`. Functionally supported but formally **candidate_uncertain**.

- **MoaA over-propagation:** PP_1969 and PP_2482 are non-canonical radical-SAM paralogs labeled "MoaA"; only PP_4597 is the canonical cyclase. Risk of double-counting step 1.
- **Broad/loose EC & domain mappings:** the MobA-like PF12804 domain is shared by MobA (MGD), MocA (MCD), and other NTP transferases — protein-name "MobA-like NTP transferase domain-containing protein" is intentionally noncommittal and should not be read as MGD.
- **Boundary error in the assigned bucket:** ppu00790 (Folate) is not the Moco pathway; the module should be curated against ppu04122 + `moeA`/`mobA`.

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6. Module and GO-curation recommendations

Per-step calls: - cPMP formation (MoaA, MoaC): **covered** — PP_4597, PP_1292. - Sulfur-carrier activation + MPT synthesis (MoeB, MoaD, MoaE): **covered** — PP_0735, PP_1293, PP_1294 (**add PP_0735 & PP_1293 to the module gene set**). - MPT adenylylation: **covered-by-alternative (MoaB lineage)**, PP_2122/PP_4600; annotate as "MogA function fulfilled by MoaB paralog." Not a gap. - Molybdate insertion (MoeA): **covered** — PP_2123. - MGD (MobA): **covered** — PP_3457. - MCD (MocA): **candidate_uncertain** — PP_2483 (primary) / PP_4230; promote for `mocA` assignment. Strain functional evidence argues for eventual **covered**. - Overall module: **satisfiable / covered** in PSEPK.

Module-document / boundary: - module_needs_revision (metadata): update the candidate-gene provenance so Moco members are gathered from ppu04122 + `moeA`/`mobA`, not the folate bucket. Exclude folate (`folB/E/C/K/P/A/M`, `pab`, `phh`), riboflavin (`rib`), and queuosine (`que`) genes from the Moco module. - The generic module boundaries are **appropriate** for this organism; only the *local bucket mapping* is wrong.

GO-curation: - Assign/confirm: PP_4597 GO:0061799 (cyclic pyranopterin monophosphate synthase / GTP 3′,8-cyclase GO:0061798), PP_1292 GO:0061799, PP_0735 GO:0008641, PP_1293/PP_1294 GO:0030366 (molybdopterin synthase), PP_2123 GO:0061604 (molybdopterin molybdotransferase), PP_3457 GO:0061603 (Mo cofactor guanylyltransferase). - **Likely GO/term request:** a **MocA / Mo cofactor cytidylyltransferase (GO:0061602)** assignment for PP_2483 (and possibly PP_4230). Consider a MoaB "molybdopterin adenylyltransferase" functional annotation (GO:0061598) for PP_2122/PP_4600 to document the MogA-surrogate role.

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7. Genes to promote to full `fetch-gene` review

1. **PP_0735** `moeB` — essential, missing from candidate list; confirm and add.
2. **PP_1293** `moaD` — essential sulfur carrier, missing from candidate list; confirm and add.
3. **PP_2483 (Q88K10)** — primary MocA/MCD candidate; resolve MGD vs MCD assignment and gene naming.
4. **PP_4230 (Q88F68)** — secondary MobA-like/MocA candidate; disambiguate from PP_2483.
5. **PP_2122** / **PP_4600** `moaB-I/II` — confirm MPT-adenylyltransferase (MogA-surrogate) capability; decide the covered-by-alternative call.
6. **PP_1969 & PP_2482 ("MoaA")** — adjudicate over-propagation; likely re-annotate as non-canonical radical-SAM paralogs.

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8. Key references

- P 12730200
  — Parschat et al. 2003. Qox (Mo-MCD quinaldine 4-oxidase) *"synthesized in a catalytically fully competent form by a heterologous host, P. putida KT2440 pKP1."* Direct KT2440 evidence for functional Mo-MCD maturation.
- P 12654012
  — Frerichs-Deeken et al. 2003. Qor (Mo-MCD) in KT2440 pUF1 is Mo-deficient — client-specific insertion limitation (outside module scope).
- P 12023088
  — Israel et al. 2002. Ior (Mo-MCD) inactive in KT2440 pIL1; accessory maturation genes required — apoenzyme loading, not Moco synthesis.
- P 41933999
  — Guo et al. 2026. Molybdenum-containing xanthine oxidase produced in KT2440; XodC accessory role in maturation.

- [P 8798497]
  — Bläse et al. 1996. Molybdopterin-cytosine-dinucleotide (MCD) cofactor in quinoline 2-oxidoreductase, functional in recombinant KT2440.
- UniProt proteome **UP000000556** (all locus/EC/InterPro annotations cited above); KEGG maps **ppu04122** (sulfur relay), **ppu00790** (folate), module **M00880**; InterPro **IPR013483** (MoaA), **IPR013482** (MobA/MGD).

Uncertainty & species-transfer notes

- Core enzyme presence/EC assignments: **strong** (UniProt/InterPro, target proteome; homology-based but high-confidence).
- MCD-branch functionality: **direct KT2440 experimental** evidence (heterologous-host maturation).
- MPT-adenylylation via MoaB and MocA identity of PP_2483/PP_4230: **inferred** from domain architecture, genomic context, and functional maturation data — no direct KT2440 enzymatic assay of these specific genes. These are the key open questions for expert review.

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